One Plate, Any Prior: Learning Decoder-Agnostic Illumination for 3D Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Computational 3D fluorescence microscopy relies on a reconstruction prior wherever the measurements are silent, and a strong prior can return a sharp volume whose depth the measurements never recorded. Depth must therefore be encoded by the optics, and an optical design is only as useful as its benefit to the decoders that will actually read it. We learn the illumination of a fixed-focus microscope once, without a decoder in the loop, and test it with four. A plane wave is scanned in angle through a fixed phase plate placed before an intermediate image plane, so that every depth receives its own sequence of excitation patterns while the specimen and the detection focus stay still. We train the plate with a specimen-free objective that decorrelates the measurement signatures of different depths. In simulations of mouse pyramidal neuron volumes at eight-fold measurement compression and equal excitation dose, the learned plate lowers the depth error of total variation, pretrained and neuron-trained plug-and-play denoisers, and a diffusion posterior by 26–63 % relative to a random plate, with cell-level confidence intervals that exclude zero. It reaches the random plate's full-scan depth accuracy with one eighth of the frames and tightens distribution-free conformal depth intervals by 27 %. An in-focus structured-illumination encoder stays at the error of a depth-blind flat estimate under every decoder: a better prior sharpens what the optics recorded but cannot create depth.

1. Introduction

Fixed-focus three-dimensional fluorescence microscopy encodes depth into planar camera images and recovers the volume computationally. The decoders that perform this recovery are changing quickly, from hand-crafted regularisers to plug-and-play denoisers and diffusion models [8, 30, 37, 38]. A stronger prior returns a cleaner volume, but where the measurements leave depth undetermined it also returns a plausible depth that the data never established [4, 10]. In neuronal imaging this is the quantity that matters: whether a dendrite lies above or below its neighbour decides which structures it can touch.

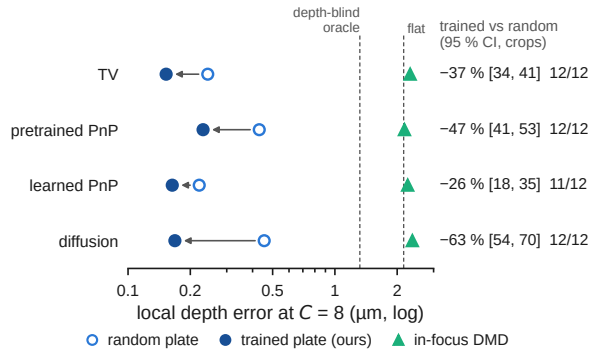


Figure 1. **One plate, any prior.** Mean local depth error of 12 mouse pyramidal neuron volumes reconstructed by four decoders from the same simulated measurements (eight-fold compression, equal excitation dose). The plate is learned once, without a decoder, and lowers the depth error of every decoder (right: relative change against the random plate it started from, cell-bootstrap 95 % interval, crops improved). An in-focus DMD encoder stays at the error of a depth-blind flat estimate under every decoder, including the neuron-trained and generative ones.

Depth information therefore has to come from the optics. Learned optics are usually optimised end-to-end with a reconstruction network [12, 23, 29], which ties the hardware to the decoder it was trained with, although the hardware is fixed once fabricated and the decoder is not. Decoder-free criteria, such as Fisher information, coherence or estimated mutual information [25, 28, 33], avoid this coupling, but whether their benefit transfers across decoders is rarely tested. We ask a direct question: can an illumination learned once, without any decoder, improve depth recovery under every decoder we can apply to it, and can that improvement be guaranteed rather than averaged?

We study this on a fixed-focus microscope that encodes depth in the excitation (Fig. 2). A plane wave is scanned in angle through one fixed phase plate placed a short distance before an intermediate image plane; a high-NA relay maps the field into the specimen, and a low-NA camera at fixed focus records one image per angle. A tilt translates the diffracted pattern by an amount that grows with propagation distance, so each depth sees its own sequence of excitation

values without axial scanning or refocusing. The phase map is the learnable element. We train it to decorrelate the measurement signatures of different depths, through a voxel-integrated vectorial model of the optics, with no specimen and no reconstruction in the loop.

We then fix the plate and decode its measurements with four priors that share nothing but the data term: total variation, a pretrained natural-image denoiser, a denoiser trained on neurons and a diffusion posterior. The learned plate lowers the depth error of all four (Fig. 1), matches the random plate's full scan with one eighth of the frames, and turns the average gain into tighter distribution-free depth intervals. An in-focus structured-illumination encoder, evaluated identically, gives no decoder any depth to work with. Our contributions are:

1. **A fixed-focus depth encoder.** Angle scanning through a defocused phase plate gives every depth its own excitation sequence; a voxel-integrated vectorial operator models it accurately and is assembled efficiently.
2. **Decoder-free illumination learning.** A specimen-free depth-decorrelation objective trains the plate. At full size it outperforms worst-case coherence and Fisher-information objectives on the same budget.
3. **Decoder-agnostic evaluation.** At equal dose, the learned plate lowers depth error by 26–63 % under four decoders, replicates with a second training seed, tightens conformal depth intervals from ± 0.65 to $\pm 0.47\,\mu\mathrm{m}$ at 90 % coverage, and keeps its gain under calibrated fabrication and alignment errors. A depth-blind encoder shows what no prior can repair.

2. Related work

Depth encoding in fluorescence microscopy. Light-field microscopy, DiffuserCam and Miniscope3D encode volumetric information in the detection response [3, 6, 33]. Excitation coding offers a complementary route. Yeh *et al*. [34] reconstruct three-dimensional fluorescence using translated speckle illumination and axial sample scanning, while Pascucci *et al*. [24] exploit axial speckle decorrelation to recover volumes from a two-dimensional scan. Yilmaz *et al*. [35] vary the incident angle through a fixed scattering layer for two-dimensional recovery, and Jiang *et al*. [14] use tilted pattern projection to produce depth-dependent lateral shifts. Our acquisition combines these mechanisms: the incident angle is scanned through a fixed, defocused phase plate, with a stationary specimen and fixed camera focus, and the plate is learned.

Learning optical encoders. End-to-end methods optimise optics jointly with a reconstruction network, for extended depth of field [29], monocular depth [7, 32], 3D localisation [23], coded illumination [17] and compressive fluorescence microscopy [12]. The learned optics are then matched to one decoder. Decoder-free design instead scores the measurements: Shechtman *et al*. [28] maximise Fisher information, Miniscope3D [33] penalises the coherence of depth-dependent point-spread functions, building on coherence minimisation in sensing design [9], and IDEAL [20, 25] optimises an estimate of mutual information. Our objective decorrelates the voxel measurement signatures of different depths, requires no specimen examples, and is compared with coherence and Fisher-information objectives under the same budget. Our main departure is the evaluation: a design is judged by its gain under several independently built decoders, rather than one.

Priors, hallucination and guarantees. Plug-and-play methods insert learned denoisers into iterative solvers [15, 30, 37], and diffusion models now provide generative priors for inverse problems [8, 16, 38]. A prior can produce structure that the forward model does not determine [4, 10, 13], and posterior spread from a learned model is not a calibrated error bound [22]. Distribution-free methods calibrate reconstruction errors on held-out data [1]; conformal risk control [2] bounds the expected value of a monotone loss. We use it to state depth placement as a guarantee, and use set-membership bounds over all measurement-consistent volumes [11, 21] to examine what the measurements alone determine.

Forward modelling. Incoherent structured-illumination models combine excitation intensity with the emission response [34], and wave-optics models account for diffraction in volumetric imaging [6]. We integrate fine-grid optical responses over reconstruction voxels, with a vectorial high-NA relay [19, 26]. Our reconstruction uses established components, adapting multi-slice fusion [18] to three-dimensional ADMM; we do not propose a new reconstruction algorithm.

3. Angle-scanned phase-encoded 3D imaging

Acquisition. We vary the illumination direction across two angular coordinates while keeping the phase plate fixed (Fig. 2). For transverse position $\mathbf{r} = (x, y)$, the incident plane wave is $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^{\top}\mathbf{r}}$, with $\mathbf{s}_t = \cos\theta_t(\cos\alpha_t, \sin\alpha_t)$, where α_t is the azimuth and θ_t parametrises the direction in the vertical plane containing that azimuth. The corresponding forward direction is $(\mathbf{s}_t, \sin\theta_t)$, and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ for the simulated refractive index of one. Thus, the two coordinates control the orientation and signed magnitude of the transverse phase gradient. The scan contains 256 distinct directions with $0.5 \le \|\mathbf{s}_t\| \le \sqrt{3}/2$. Each incident field passes through the phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagates a distance D to the intermediate illumination image plane by angular-spectrum propagation. The NA-0.8 objective pupil filters the modulated field

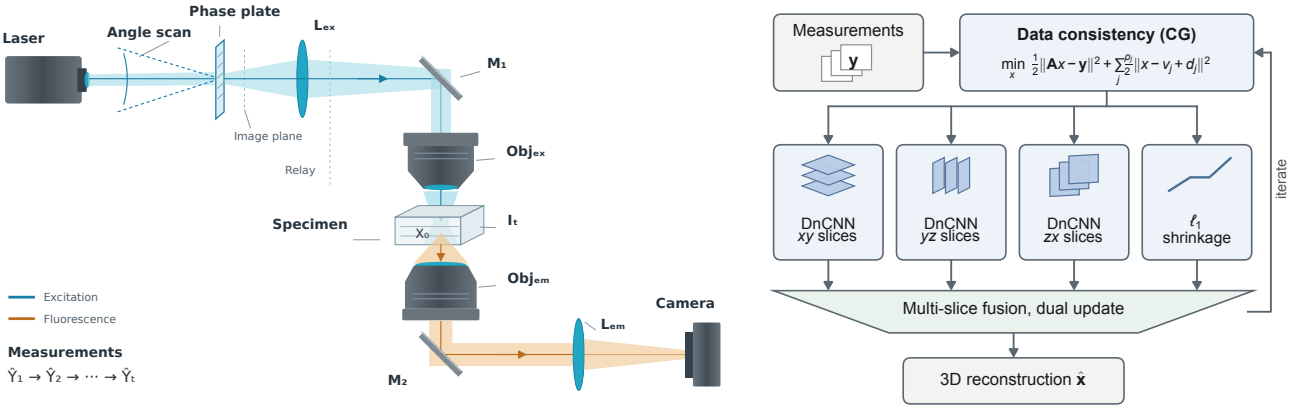


Figure 2. **Fixed-focus acquisition and reconstruction.** Left: a tilted plane wave illuminates a fixed phase plate placed 6.6 μm (specimen-space units) before an intermediate illumination image plane; an NA-0.8 vectorial relay maps the field into the specimen; a camera at a fixed detection focus records one image per illumination angle. Only the input angle changes: plate, specimen and focus are stationary, and nothing is scanned axially. Blue: excitation; orange: fluorescence. Right: one of the four decoders of Sec. 4, multi-slice plug-and-play recovery, alternates a conjugate-gradient data-consistency step with a 2D denoiser applied to the xy, yz and zx slices of a shared volume and an ℓ_1 shrinkage, fused by ADMM.

before a vectorial Richards–Wolf relay [19, 26] produces the specimen excitation $I_t = \sum_{c\in\{x,y,z\}} |E_{t,c}|^2$, assuming x-polarised illumination. Coordinates and distances are expressed in specimen-equivalent units. The camera records one fluorescence image per direction at a fixed detection focus. Acquisition is sequential, not single-shot.

Why the angle encodes depth. Propagation converts the plate's phase structure into spatially varying excitation. Scanning the incident angle shifts this pattern differently at different depths, giving each location a depth-dependent illumination sequence. An ideal scalar Fresnel model illustrates the mechanism: without pupil clipping, an angular change $\Delta\mathbf{s}$ displaces the pattern by $(D+z)\Delta\mathbf{s}$, where $D+z$ is the distance from the plate. Different depths therefore translate at different rates, while plate defocus positions the shear pivot outside the specimen. This relation provides intuition; our high-NA vectorial model captures the finite pupil and departures from pure translation.

Image formation. Let $X_0(\mathbf{r}, z)$ denote the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z')X_0(\mathbf{r}', z')I_t(\mathbf{r}', z')\,\mathrm{d}^2\mathbf{r}'\,\mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity point-spread function and Ω is the specimen volume. Pixel p, occupying region D_p, records the normalised sample $\hat{Y}^0_{t,p} = |D_p|^{-1}\int_{D_p} g_t(\mathbf{r})\,\mathrm{d}^2\mathbf{r}$. Fluorescence contributions therefore add incoherently before pixel averaging. Multiplication by the excitation pattern mixes specimen spatial frequencies before detection blur, so the camera need not resolve the illumination pattern itself. Depth discrimination depends on how these measurements vary across illumination angles.

Voxel-integrated operator. To capture excitation variations within reconstruction voxels, we evaluate the optics on a finer grid and integrate the measurement responses over each voxel. Let B be the fine-grid measurement operator, including emission propagation and pixel averaging, and let U replicate each reconstruction coefficient over its fine-grid cells. Then $\mathsf{A} = \mathsf{BU}$ preserves the fine-grid response for every specimen represented in this piecewise-constant basis.

Sub-voxel specimen structure remains unrepresented. For a fine-grid specimen $\mathbf{x}_f$, its coarse representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, obtained by voxel averaging, satisfies

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{URx}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{2}$$

Thus, integration retains within-voxel optical variation, while $\boldsymbol{\delta}_{\mathrm{disc}}$ captures specimen approximation error.

Linear, incoherent fluorescence enables efficient assembly:

$$\mathsf{A}_{(t,p),j} = \sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{3}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells, $(\mathbf{r}_i, z_i)$ is the location of cell i, and $g_{p,i}$ is its emission footprint at pixel p, including its integration weight. These footprints are computed once and reused across angles. After precomputation, direct assembly costs $O(TPF)$, compared with

$O(TNF \log F)$ for constructing columns through separate FFT-based forward simulations, with T patterns, P detector samples, F fine-grid cells and N reconstruction voxels.

Learning the plate. We optimise the plate phase ϕ to reduce correlations between measurement signatures at different depths, with the aim of improving depth discrimination. For a voxel at lateral position $\mathbf{r}$ and depth z, let $\mathbf{a}_{\mathbf{r},z}$ denote its centred, unit-norm operator column (Eq. (3)), collecting detector responses across illumination angles. We minimise

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r}} \sum_{z \neq z'} w_{zz'} \left(\mathbf{a}_{\mathbf{r},z}^{\top} \mathbf{a}_{\mathbf{r},z'}\right)^2, \tag{4}$$

where mirror-depth pairs about the detection focus receive weight two and other pairs receive weight one. Squaring penalises correlations of either sign; normalisation prevents uniform amplification of the signatures from reducing the loss.

Training differentiates through the excitation, vectorial relay and detection model, without specimen examples or a reconstruction solver. The objective is a surrogate for depth discrimination, not a guarantee of stable recovery; whether it helps is decided by the decoders of Sec. 4, none of which the plate has seen. Training details are given in the supplement.

4. Decoders and depth guarantees

The plate is fixed before any decoder is built. We decode its measurements with four priors from different families. All of them are trained or chosen without measurements, so the same prior serves every encoder; only scalar weights are selected per encoder and compression, on validation volumes.

Total variation. Box-constrained least squares with isotropic 3D total variation [27], solved by ADMM [5]: a non-learned regularity prior.

Multi-slice plug-and-play. We combine least-squares data fitting with multi-slice denoising and sparsity through plug-and-play ADMM [18, 30]. The branches $\mathcal{J} = \{xy, yz, zx, \mathrm{sp}\}$ comprise three slice-denoising branches and one sparsity branch. For orientation o, P_o rearranges the shared volume into slices, which a 2D DnCNN $\mathcal{D}$ [36] denoises independently, giving the volume denoiser $\mathcal{D}_o = \mathsf{P}_o^{-1} \mathcal{D} \mathsf{P}_o$. Each iteration solves

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \textstyle\sum_{j \in \mathcal{J}} \frac{\rho_j}{2} \|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2 \tag{5}$$

by conjugate gradients, followed by $\mathbf{v}_o^{k+1} = \mathcal{D}_o(\mathbf{x}^{k+1} + \mathbf{d}_o^k)$, soft thresholding $\mathbf{v}_{\mathrm{sp}}^{k+1} = \mathcal{S}_{\lambda/\rho_{\mathrm{sp}}}(\mathbf{x}^{k+1} + \mathbf{d}_{\mathrm{sp}}^k)$ and dual updates $\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}$. We use two versions. The *pretrained* one uses a DnCNN trained on natural images. The *learned* one replaces it with neuron-trained denoisers, a 2D network shared by the three orientations and a 3D network that replaces all three slice branches [15, 37], trained for blind Gaussian denoising on volumes of an animal excluded from testing.

Diffusion posterior. A 3D U-Net denoiser in the EDM parametrisation [16] is trained on the same volumes. We sample with DiffPIR [38]: at each of 100 noise levels, the denoised estimate is pulled towards the data by an exact proximal step on the Poisson-weighted least-squares term $\frac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_{\mathsf{W}}^2$, solved by conjugate gradients, with W the inverse photon-noise variance. We report the mean of eight posterior samples. This decoder shares only the operator with the others.

Conformal depth intervals. An average error is not a statement about a new specimen. For each lateral column c of a reconstruction with signal, let $\hat{z}_c$ be its axial intensity centroid and z_c the true one. We report the interval $\hat{z}_c \pm \lambda$, with loss $L(\lambda)$, the signal-weighted fraction of columns whose true centroid falls outside it. L is monotone in λ and bounded by one, so conformal risk control [2] applies: with empirical risk $\hat{R}_n(\lambda)$ on n calibration volumes reconstructed exactly like the test volumes,

$$\hat{\lambda} = \inf\left\{\lambda : \tfrac{n}{n+1}\hat{R}_n(\lambda) + \tfrac{1}{n+1} \leq \alpha\right\} \tag{6}$$

gives $\mathbb{E}[L(\hat{\lambda})] \leq \alpha$ for a new volume exchangeable with the calibration set. The guarantee holds for any decoder; a better encoder can only make $\hat{\lambda}$ smaller.

What the measurements alone determine. Priors aside, the data define a measurement-consistency set $\mathcal{C}(\mathbf{y}) = \{\mathbf{x} \in [0,1]^N : \ell_i \leq (\mathsf{A}\mathbf{x})_i \leq h_i\}$, where the per-row intervals contain the noise-free measurement with stated joint probability. Per-voxel bounds $\min_{\mathcal{C}} x_v$ and $\max_{\mathcal{C}} x_v$ (two linear programs per voxel) show how far any consistent volume can move. The plate objective acts on these bounds: for two voxels with unit-norm columns of correlation ρ in an ℓ_2 ball of radius ε, the range of either voxel is $\pm\varepsilon/\sqrt{1-\rho^2}$, from the Schur complement of the 2×2 Gram matrix. Lowering depth correlations therefore narrows the set; we use this as intuition, not a theorem, and measure it on a reduced field where exact optimisation is feasible.

5. Experimental setup

All experiments are simulations with the optics of Sec. 3: 532 nm excitation and detection, excitation NA 0.8, detection NA 0.1, a phase plate with 0.32 μm features at $D = 6.6\,\mu$m, 256 directions, 0.08 μm optical and 0.32 μm analysis grids, 0.64 μm reconstruction voxels, and linear fluorescence in a homogeneous non-scattering medium.

Volumes and compression. The reconstruction grid is $16 \times 48 \times 48$ ($N = 36{,}864$, $30.7 \times 30.7 \times 10.2\,\mu\mathrm{m}^3$) with 12×12 camera samples per image, so $M = 144k$ for k directions and the compression is $C = N/M \in \{1, 4, 8\}$ for $k = 256, 64, 32$ (five random direction subsets for $C > 1$). Reference volumes are two-photon stacks of mouse pyramidal neurons from the public DEEP^2 data [31], min–max normalised and resampled to the analysis grid. The test set holds two prespecified crops from each of six cells of three animals. The neuron-trained denoisers and the diffusion model are trained on 459 crops of a fourth animal; held-out crops of that animal, simulated like the test crops, calibrate the conformal intervals (35 non-overlapping crops; all 240 crops as a sensitivity check). Scalar weights of every decoder are selected per encoder and compression on two vascular crops of a separate stack, on grids that extend automatically at their edges. No test volume is used for training, tuning or calibration.

Encoders and photon budget. The random plate (three realisations), a defocused binary amplitude mask at the plate's position, the same scan without a plate, and an in-focus DMD with 256 random binary patterns share relay, pupil, detector and assembly. Their excitation is simulated on an optical grid larger than the field, so beam edges lie outside it, and normalised to the same deposited dose. Counts are $\mathrm{Poisson}(\kappa_0\, 256\, y_i/k)$: a fixed total budget shared by the k frames, with κ_0 set per specimen so that the random plate reaches 15 dB at $k = 256$ (1000 photons per pixel per frame at the mean signal). The learned plate starts from random realisation 0 and changes only inside the specimen field; it is trained with two seeds.

Metrics and statistics. Besides PSNR we report foreground SSIM and the *local depth error*: the mean absolute difference between the true and reconstructed axial centroids of each lateral column, weighted by the true column brightness over columns above 10 % of the brightest. Two references calibrate it: a depth-blind oracle with the exact lateral structure and each column spread over the crop's mean axial profile scores $1.32\,\mu\mathrm{m}$, and a flat estimate, uniform in depth, $2.15\,\mu\mathrm{m}$. Crops of one cell are not independent, so intervals resample cells (six cells, two crops each, 10^4 resamples) and gains are paired on identical measurements.

Reduced field. Exact bounds over $\mathcal{C}(\mathbf{y})$ use dense linear programs, so they run on a reduced field of the same optics ($4 \times 12 \times 12$ voxels, 3×3 pixels, $k = 32$) with Bonferroni intervals at $\alpha = 0.05$ and Gaussian noise at 15 dB, on 24 synthetic phantoms with three noise seeds and on 48 windows cut from the test neurons (supplement).

		multi-slice PnP			box-constrained TV			beads (PnP)	
C	encoder	PSNR	FG-SSIM	depth	PSNR	FG-SSIM	depth	depth	ghost
1	phase plate	35.73	0.814	0.357	38.15	0.881	0.187	0.00	0.033
	amp. mask	35.24	0.788	0.347	37.59	0.857	0.191	0.00	0.030
	no plate‡	30.87	0.538	0.753	32.78	0.617	0.477	0.32	0.363
	in-focus DMD	28.96	0.501	2.168	28.48	0.435	2.285	3.84	1.007
4	phase plate†	35.40	0.801	0.389	37.34	0.852	0.223	0.01	0.013
	amp. mask†	34.88	0.778	0.382	36.79	0.823	0.239	0.01	0.017
	no plate‡	30.41	0.513	1.036	32.09	0.580	0.706	0.34	0.452
	in-focus DMD	28.91	0.493	2.168	28.49	0.465	2.326	2.94	1.013
8	phase plate†	34.86	0.784	0.430	37.00	0.847	0.243	0.01	0.039
	amp. mask†	34.14	0.744	0.446	36.74	0.827	0.237	0.08	0.048
	no plate‡	30.18	0.494	1.102	31.81	0.579	0.737	0.49	0.335
	in-focus DMD	28.83	0.482	2.169	28.53	0.473	2.311	2.60	1.013

Table 1. **Encoder comparison** (mean over 12 test crops; depth errors in μm; beads: depth error of detected beads and mirror-ghost energy ratio). †PnP: mean over three plate or mask realisations; TV and all other rows: realisation 0. ‡Simulated aperture four times the field (two times for all other rows; the plate changes by ≤ 0.07 dB). Equal deposited excitation dose. Depth-blind oracle $1.32\,\mu\mathrm{m}$, flat estimate $2.15\,\mu\mathrm{m}$.

6. Results

6.1. Defocus under the angle scan encodes depth

We first ask whether the untrained mechanism works. Figure 3 and Tab. 1 compare the random plate with three encoders that share its relay, detector, dose, volumes, decoders and tuning protocol. As predicted (Sec. 3), the plate's excitation shears with tilt at 0.7–0.9 of the thin-screen rate, per-pixel excitation sequences decorrelate within about $1\,\mu\mathrm{m}$ of depth, and point emitters are assigned a wrong depth in 1.5–1.8 % of cases (three plates) against 4.4–5.3 % (three defocused masks), 25 % (DMD, half at the mirror depth) and 89 % without the plate. The in-focus DMD stays at the depth error of a flat estimate (2.17 against $2.15\,\mu\mathrm{m}$) while the plate reaches $0.36\,\mu\mathrm{m}$ (pretrained prior, $C = 1$). Without the plate, depth error rises towards the depth-blind oracle as the simulated aperture grows and its edge stops carrying depth (supplement). The defocused amplitude mask comes close to the plate: over three realisations of each, the plate leads in PSNR in all nine pairings ($+0.4$ to $+1.2$ dB at $C = 8$) but not in depth error. Depth is therefore encoded by defocus under the angle scan, and a random plate is already a strong encoder; the question is how much learning adds to it.

6.2. One plate, any prior

We now fix the learned plate and decode its measurements with each prior of Sec. 4 (Fig. 1 and Tab. 2). At $C = 8$ it lowers the depth error by 37 % with total variation, 47 % with the pretrained denoiser, 26 % with the neuron-trained denoiser and 63 % with the diffusion posterior. Every interval excludes zero, depth improves on 11–12 of 12 crops, and PSNR rises by 1.3–4.4 dB. A second training seed gives -29 to -64 %. The gain holds at every compression we tested

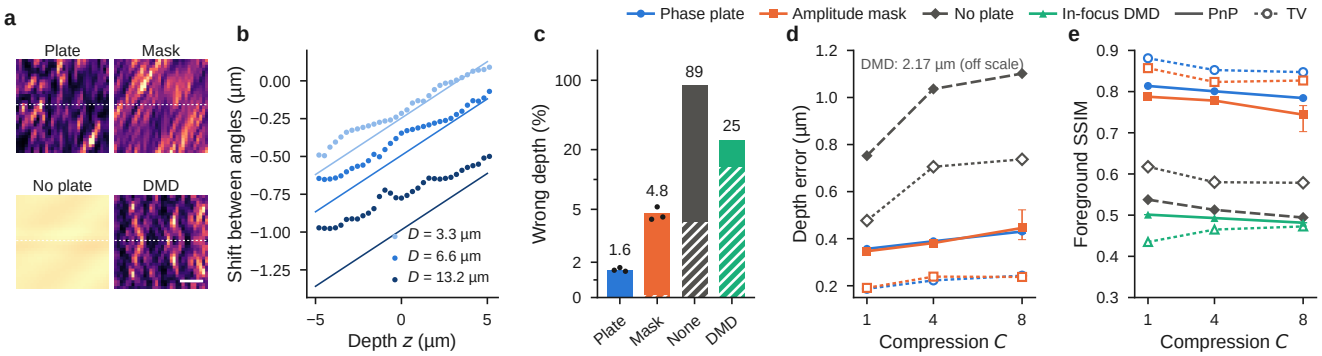


Figure 3. **Defocus under the angle scan encodes depth; in-focus patterns and a plain scan do not** (full-size field, equal deposited excitation dose, overfilled illumination aperture). (a) One excitation pattern per encoder, xz section through the specimen depth ($\pm 5\,\mu$m, focus dotted), each tile scaled to its own 99.5th percentile. (b) Lateral shift of the plate's excitation between two scan angles at every depth for plate defocus $D = 3.3$, 6.6 and 13.2 μm; lines: thin-screen prediction, sheared about the plate's conjugate plane at distance D. (c) Point emitters assigned a wrong depth by maximum likelihood with unknown brightness at $k = 32$ (dots: three plate and three mask realisations; hatched: mirror depth). (d, e) Reconstructions of 12 neuron crops from 6 cells: local depth error and foreground SSIM versus compression, multi-slice PnP (solid, no plate dashed; plate and mask at $C = 4, 8$: mean and range over three realisations) and box-constrained TV (dashed); the DMD's depth error of 2.17 μm is off scale. No plate: simulated aperture four times the field in (a, d, e), twice in (c).

	random plate		learned plate		depth change vs random		DMD
decoder	PSNR	depth	PSNR	depth	seed 0 [95 % CI]	seed 1	depth
TV	37.00	0.243	38.61	0.153	−37 % [−41, −34]	−38 %	2.31
pretrained PnP	34.81	0.431	36.15	0.230	−47 % [−53, −41]	−46 %	2.17
learned PnP	37.11	0.221	38.83	0.164	−26 % [−35, −18]	−29 %	2.25
diffusion	34.34	0.455	38.74	0.168	−63 % [−70, −54]	−64 %	2.37

Table 2. **The learned plate under four decoders** ($C = 8$, mean over 12 test crops × 5 measurement draws, equal dose; PSNR in dB, local depth error in μm). Depth change: learned against random plate on identical measurements, cell-bootstrap 95 % interval; the plate is trained once with each of two seeds.

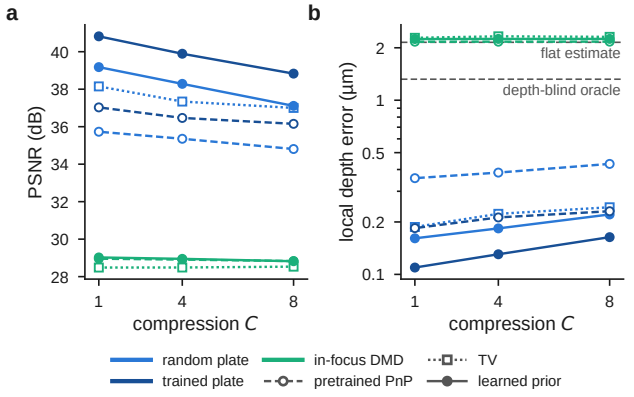


Figure 4. **Optics and prior gains add, where the optics record depth** (full-size field, 12 test crops, equal deposited dose). PSNR (a) and local depth error (b) against compression C; colour: encoder, line: decoder. Dashed lines in (b): the depth-blind oracle and the flat estimate. No decoder moves the DMD off the flat estimate.

(Fig. 4): −45 to −48 % with the pretrained denoiser, −25 to −37 % with total variation and −26 to −32 % with the learned denoiser at $C = 1$, 4 and 8.

Eight times fewer frames. At the same total dose the learned plate with 32 frames matches the random plate with all 256: with the learned denoiser 0.164 against 0.161 μm (difference −2 % [−17, +15]), with total variation 0.153 against 0.187 μm, and with the pretrained denoiser it is better (−35 % [−42, −28]).

The decoders converge. With the random plate the four decoders disagree: their depth errors span 0.22–0.46 μm, and the diffusion posterior trails the learned denoiser by 2.8 dB. With the learned plate they span 0.15–0.23 μm, and the two neuron-trained priors land within 0.1 dB and 0.004 μm of each other. When the measurements record depth, the choice of prior matters less. We do not claim the diffusion prior as a better decoder: on the random plate it is clearly weaker than the learned denoiser, which inflates its relative gain, and its posterior spread is over-confident (supplement).

6.3. Why depth decorrelation

Table 3 trains the plate with three alternative decoder-free objectives from the same starting plate, sampling and step budget: the worst-case coherence of depth columns, as in Miniscope3D [33], a Poisson Fisher-information objective on depth with unknown brightness, in the spirit of [28], and our loss without centring. All improve on the random plate. Coherence and Fisher information keep 50–63 % of

	$C = 1$		$C = 8$	
plate objective	PSNR	depth	PSNR	depth
none (random plate)	39.18	0.161	37.11	0.221
worst-case coherence	39.95	0.135	38.20	0.186
depth Fisher information	40.15	0.128	38.15	0.190
depth decorrelation, uncentred	40.86	0.108	38.85	0.164
depth decorrelation (ours)	40.82	0.109	38.83	0.164

Table 3. **Plate objectives** under the same start, budget and seed, decoded with the learned denoiser (12 test crops; depth in μm). Each plate has its own validation tuning.

	conformal depth interval ($\alpha = 0.1$, $C = 8$)			bound width
plate	$\pm\hat{\lambda}$ (μm)	vs random [95 % CI]	test miss	reduced field
random	0.646	–	0.060	1
learned, seed 0	0.474	-26.7 % [-30.1, -22.9]	0.075	0.79
learned, seed 1	0.486	-24.8 % [-29.6, -21.6]	0.060	0.79

Table 4. **Depth guarantees.** Conformal intervals of the learned-denoiser reconstructions, calibrated on 35 held-out crops (calibration-bootstrap 95 % interval) and checked on the 12 test crops $\times$ 5 draws (test miss: signal-weighted miscoverage, target ≤ 0.1). Bound width: mean per-voxel width over all measurement-consistent volumes on the reduced field (15 dB, 72 queries), relative to the random plate.

our depth gain over it, and centring makes no difference. Penalising every depth pair at every voxel serves this encoder better than optimising the worst pair or a local information measure.

6.4. From average gains to guarantees

Conformal depth intervals. Calibrated on held-out crops, the conformal intervals of Eq. (6) hold on the test cells of other animals (Tab. 4; miscoverage 0.06–0.075 against $\alpha = 0.1$). With 90 % guaranteed coverage a column's depth lies within $\pm 0.47\,\mu$m with the learned plate and within $\pm 0.65\,\mu$m with the random plate, 27 % tighter (second seed: 25 %; calibrated on all 240 crops: 26 %). The average gain of Sec. 6.2 thus becomes a bound that holds, in expectation, for a new volume, without a model of the reconstruction errors.

What the measurements alone determine. Without any prior, the per-voxel bounds over $\mathcal{C}(\mathbf{y})$ are 21 % narrower with the learned plate on the reduced field. The two-column rule of Sec. 4 predicts this from the drop in mean depth correlation $|\rho|$ from 0.868 to 0.761 ($0.80\times$ predicted, $0.79\times$ measured). Over 12.5–25 dB the narrowing equals 1.9 dB of SNR, about 2.4 times the photons. On sparse synthetic structures this lets the measurements alone certify a connection in 12 rather than 3 of 48 connected queries, gaining 9 and losing none at 15 dB (exact sign test $p = 0.004$; second seed $+5/-0$). On 48 windows cut from the test neurons,

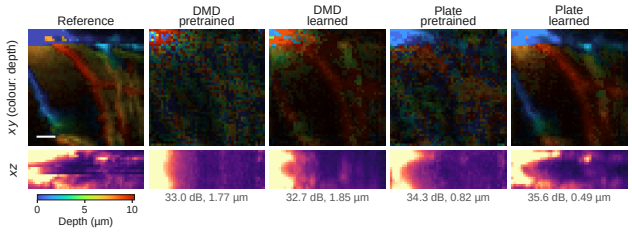


Figure 5. **A better prior, not a better answer** ($C = 8$, prespecified display crop, draw 0, identical display). Top: lateral maximum-intensity projection, colour coding the depth of the brightest voxel; bottom: xz projection; below: PSNR and local depth error. The learned prior sharpens the DMD's lateral structure but cannot give it depth; with the plate it recovers the depth layering. Bar: 5 μm.

however, no voxel of any query window is determined by any plate at 15–45 dB: at eight-fold compression dense neuronal tissue is underdetermined without a prior, whatever the plate. Guarantees about neuronal structure must therefore come through a calibrated prior, as above, and we limit them to depth placement (Sec. 7).

6.5. A prior cannot create depth

The DMD shows the other side of decoder agnosticism. Under all four decoders its depth error stays at 2.17–2.37 μm, the error of a flat estimate (Fig. 1 and Tab. 2), and a second, independent DMD pattern set reproduces every decoder to within 0.06 dB and 0.07 μm. Splitting each column's error into its mean over depth (lateral error) and the zero-mean remainder (depth error) shows why. At $C = 8$ the learned denoiser shrinks the plate's lateral error to $0.48\times$ and its depth error to $0.71\times$ (median over crops; lower on 12 of 12), but for the DMD it shrinks the lateral error to $0.77\times$ and leaves the depth error at $1.04\times$, which is 99 % of the DMD's total. Figure 5 shows the consequence: with the learned prior the DMD's dendrites look sharper in xy, and its xz section remains a depth-less smear. A prior refines what the optics recorded; it does not supply what they did not.

6.6. Calibration and tolerance

A learned plate is useful only if its gain survives fabrication and alignment. With the operator rebuilt from the fabricated plate, as a computational microscope is calibrated anyway, rounding the phase to four levels or adding 0.5 rad RMS height error to an 8-level plate keeps a depth reduction of 25.5 and 25.2 % against 25.9 % for the nominal plate (learned denoiser, $C = 8$). A plate 0.5 μm closer or farther than designed keeps 27.8 and 26.4 % after recalibration, and scan-pointing errors up to 0.01 in direction cosine keep 18–22 % without recalibration. Plate-distance and lateral plate errors left uncalibrated shift the excitation code rigidly (correlation ≥ 0.97 after the offset); they cost both plates 0.15–0.41 μm and remove the learned plate's advantage, so they must be

registered, *e.g.* with one bead. Simulating the optics at the reconstruction spacing instead of integrating over voxels raises the forward discrepancy from 1.07 to 25.7 % (supplement).

7. Discussion and limitations

Why learn the optics without a decoder. The plate is fabricated once; the decoder will be replaced as priors improve. A plate learned end-to-end with one network can exploit that network's habits, and its benefit to the next decoder is unknown. Our plate never saw a decoder, yet it helped a regulariser, two denoisers and a generative model, and made them agree more closely with one another. We did not compare against end-to-end design, which answers a different question: the best pair of optics and decoder, rather than optics that serve any decoder. Our comparison is with decoder-free objectives under the same budget (Tab. 3).

What the gain is and is not. The learned plate improves depth placement, and its conformal intervals make that improvement a guarantee. It does not make neuronal topology identifiable. At eight-fold compression no plate determines dense neuronal windows from the measurements alone (Sec. 6.4). In supplementary experiments, conformal connectivity calls on these reconstructions did not reach their nominal error rate, the diffusion posterior was over-confident (two-standard-deviation coverage 0.71–0.75 per voxel) and did not reduce confident wrong connectivity calls, and query-adaptive direction selection did not meet its prespecified bar. Connections drawn from such reconstructions remain the prior's choice.

Limitations. All results are simulations, without hardware validation, of linear fluorescence in a homogeneous non-scattering medium with an ideal relay and shot noise only; scattering, aberrations and background will reduce every encoder's depth information. The test set has 12 crops from six cells of three animals, which are not independent and from which we report cell-level intervals; the plate is trained with two seeds and the priors on one animal. The conformal guarantee assumes the calibration crops are exchangeable with new specimens; it held on test cells of other animals, but a different tissue would need its own calibration. Plate distance and lateral position must be registered before use.

8. Conclusion

The optics decide how much depth the measurements contain; the prior can only use it. A phase plate learned once, with a specimen-free depth-decorrelation objective and no decoder in the loop, lowered the depth error of a regulariser, two denoisers and a diffusion posterior, matched a full scan with one eighth of the frames and tightened distribution-free depth intervals, while an in-focus encoder left every decoder depth-blind. Learned optics should be judged by their benefit to any decoder, not only to the one they were trained with.

LLM Usage Statement

LLMs were used for grammar and fact checking.

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